

# COMP 5350/6350 Hands-On #3 Report

**Student Name:** Eristen Forsythe

## 1. Executive Summary

This investigation examined the network packet capture file `FTPPackets.pcap` to identify login activity and file transfers on the FTP server. The analysis showed that the authorized user **ftpuser** logged in and downloaded the file **employee\_details.txt**, which contained employee usernames and passwords.

After this file was downloaded, additional logins were observed from the accounts **johnsmith** and **janedoe**, indicating that the credentials from the file were used to access the server. During this several files were uploaded and downloaded, including images, PDFs, and a ZIP archive. All transferred files were successfully reconstructed from the packet capture and verified using SHA-256 and MD5 hashes.

The recovered files included financial documents, check images, a mortgage template, and an archive containing identity-related images such as a social security card and birth certificate. These materials suggest that the unauthorized users were collecting or exchanging documents that could be used for identity theft or financial fraud.

Overall, the investigation shows that exposure of plaintext credentials in `employee_details.txt` enabled unauthorized access to the FTP server. The compromised accounts were then used to transfer several files associated with personal identity and financial documentation..

The timeline illustrates how the exposure of plaintext credentials enabled unauthorized access to the FTP server, which was then used to exchange financial and identity-related documents that could potentially be used for identity theft or financial fraud.

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### Forensic Attack Timeline

| Stage | Timestamp              | User      | Source IP      | Activity Description                                                                                                                                                                   | Evidence                                                                                                                 |
|-------|------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1     | 15-11-2024<br>13:48:51 | ftpuser   | 172.19.98.16   | Authorized employee logs into the FTP server using valid credentials. This appears to be the legitimate initial access to the system.                                                  | FTP USER ftpuser and PASS commands followed by server response <b>230 Login Successful</b>                               |
| 2     | Shortly<br>after login | ftpuser   | 172.19.98.16   | The user downloads <b>employee_details.txt</b> , which contains employee usernames, passwords, and personal information. This file exposes valid FTP credentials for other accounts.   | FTP command RETR employee_details.txt with responses <b>150 Opening data connection</b> and <b>226 Transfer complete</b> |
| 3     | 15-11-2024<br>13:52:36 | johnsmith | 172.19.201.228 | A new login occurs using the account <b>johnsmith</b> shortly after the credential file was downloaded, indicating that the exposed credentials were used to gain unauthorized access. | FTP USER johnsmith followed by successful authentication response <b>230 Login Successful</b>                            |
| 4     | After login            | johnsmith | 172.19.201.228 | The unauthorized user uploads a completed check image and downloads a bank statement template, suggesting preparation or modification of financial documents.                          | FTP STOR Check1.jpg and RETR Statement2.pdf commands                                                                     |
| 5     | 15-11-2024<br>13:54:16 | janedoe   | 131.204.27.177 | Another account contained in the credential file ( <b>janedoe</b> ) logs in from a different IP address, indicating additional unauthorized access using exposed credentials.          | FTP USER janedoe followed by server response <b>230 Login Successful</b>                                                 |
| 6     | After login            | janedoe   | 131.204.27.177 | The user downloads multiple files including a check template, bank statement, and a ZIP archive containing identity documents such as an SSN card and birth certificate.               | FTP RETR check1.jpg, RETR Statement1.pdf, and RETR Documents.zip                                                         |
| 7     | Same<br>session        | janedoe   | 131.204.27.177 | The attacker uploads a <b>Mortgage.pdf</b> template, which could be used alongside the collected financial and identity documents for fraudulent financial activity.                   | FTP STOR Mortgage.pdf observed in the packet capture                                                                     |
| 8     | Final stage            | Multiple  | Multiple       | The activity results in the collection and exchange of financial documents and identity records, indicating potential preparation for identity theft or financial fraud.               | Recovered files include checks, bank statements, mortgage forms, and identity documents                                  |

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## 2. Investigation Methodology (General Overview)

The investigation was performed by analyzing a network packet capture file (**FTPPackets.pcap**) that contained recorded FTP traffic between several client machines and an FTP server. Before beginning the analysis, **hash values** were generated for the **PCAP** file to verify the integrity of the evidence.

The packet capture was examined using **Wireshark** on a **Kali Linux system**. Wireshark was used because it allows inspection of network packets, filtering by protocol, reconstruction of TCP streams, and exporting transferred files.

**Manual analysis** was performed by applying filters to isolate FTP traffic and identify authentication events and file transfers. Login attempts were identified by examining the FTP USER and PASS commands, and successful logins were confirmed using the FTP response code 230 (Login successful).

**File transfers** were identified by analyzing FTP commands such as RETR (download) and STOR (upload). These commands indicate when files are retrieved from or stored on the FTP server. The associated TCP streams were followed in Wireshark to reconstruct and export the transferred files.

**Recovered files** were saved locally and verified using hashing tools available in Kali Linux. The commands **sha256sum** and **md5sum** were used to generate hash values for each recovered file to verify file integrity.

Throughout the investigation, screenshots were captured to **document** key steps of the analysis process, including protocol statistics, login activity, FTP commands, and file recovery. These screenshots were annotated and included in the report to support the findings and ensure that the analysis process is transparent and repeatable.

Evidence file hashes were calculated prior to analysis:

MD5: 888af7dede8e3711d12eaafaf3881562

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SHA-1: 96bb76d0151c0c2b0079b1f2b77e1ca31ab8c757  
SHA-256: d2692f0fa84546a48c5b26486f3ff0718b6bf7f9f57b0776b62116cac972d1cf

## 3. Evidence Overview

### 3.1 PCAP File Properties

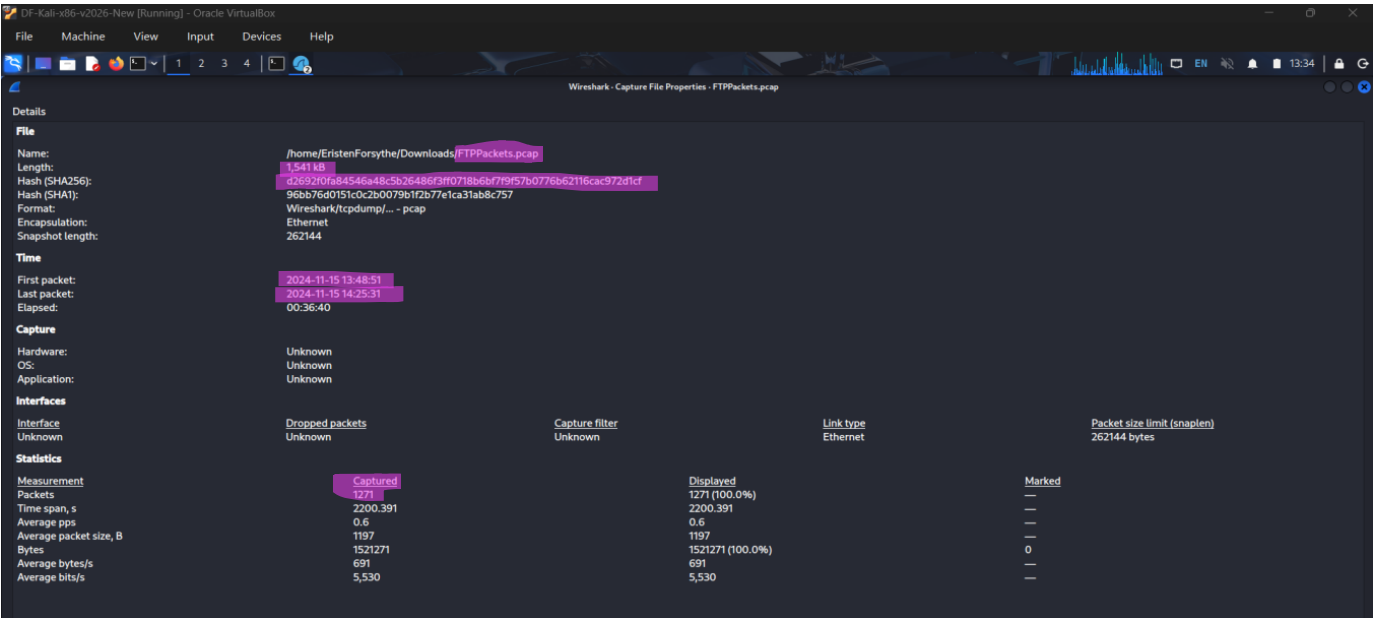
- File name: FTPPackets.pcap
- File size: 1,541 KB (1,541,631 bytes)
- Hash values (MD5/SHA-1/SHA-256)

MD5: 888af7dede8e3711d12eaafaf3881562

SHA-1: 96bb76d0151c0c2b0079b1f2b77e1ca31ab8c757

SHA-256: d2692f0fa84546a48c5b26486f3ff0718b6bf7f9f57b0776b62116cac972d1cf

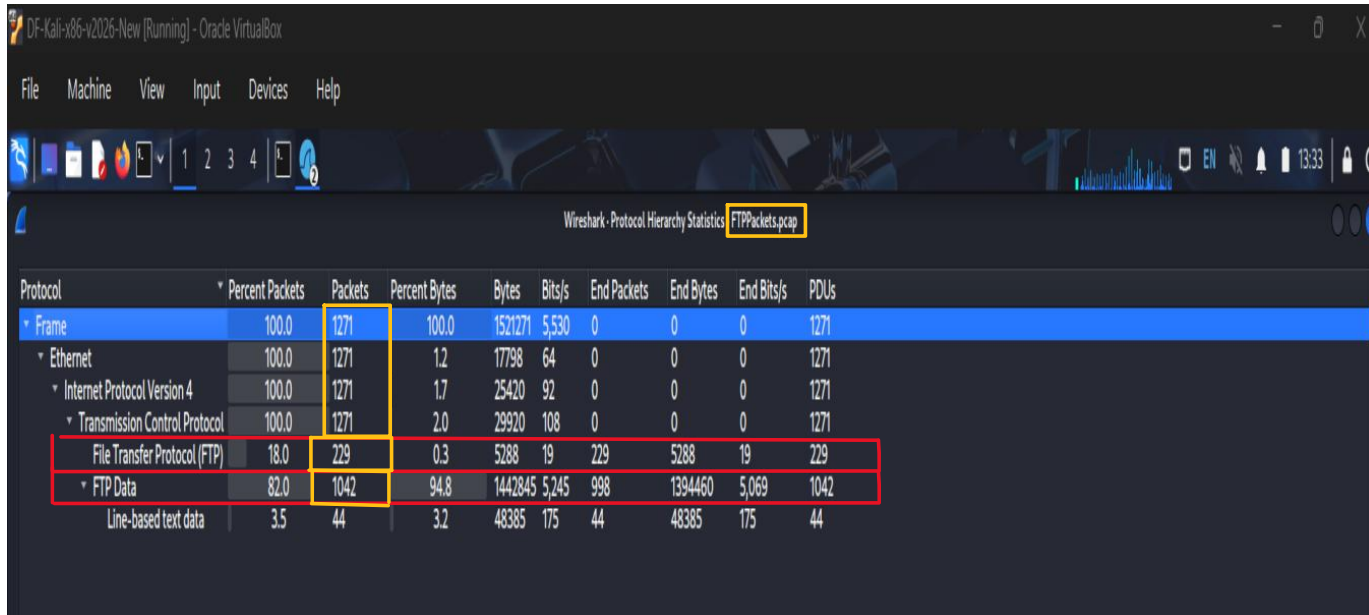
- Total packets : 1271
- Protocol breakdown: FTP control and FTP data traffic
- Capture duration: 00:36:40



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Wireshark capture file properties showing the evidence file name, capture timestamps, packet count, and SHA-256 hash used to check integrity.



Wireshark - Protocol Hierarchy Statistics: FTP.packets.pcap

| Protocol                        | Percent Packets | Packets | Percent Bytes | Bytes   | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---------------------------------|-----------------|---------|---------------|---------|--------|-------------|-----------|------------|------|
| * Frame                         | 100.0           | 1271    | 100.0         | 1521271 | 5,530  | 0           | 0         | 0          | 1271 |
| * Ethernet                      | 100.0           | 1271    | 1.2           | 17798   | 64     | 0           | 0         | 0          | 1271 |
| * Internet Protocol Version 4   | 100.0           | 1271    | 1.7           | 25420   | 92     | 0           | 0         | 0          | 1271 |
| * Transmission Control Protocol | 100.0           | 1271    | 2.0           | 29920   | 108    | 0           | 0         | 0          | 1271 |
| File Transfer Protocol (FTP)    | 18.0            | 229     | 0.3           | 5288    | 19     | 229         | 5288      | 19         | 229  |
| * FTP Data                      | 82.0            | 1042    | 94.8          | 1442845 | 5,245  | 998         | 1394460   | 5,069      | 1042 |
| Line-based text data            | 3.5             | 44      | 3.2           | 48385   | 175    | 44          | 48385     | 175        | 44   |

Protocol hierarchy statistics showing FTP control traffic and FTP data transfers present in the packet capture

### 3.2 Network Environment Summary

- FTP Server IP address: 131.204.27.176
- Server MAC address: 1c:69:7a:9c:f8:6c
- Subnet involved: 172.19.0.0/16 and 131.204.27.0/24
- Observed client IP addresses: 172.19.98.16, 172.19.201.228, and 131.204.27.177
- Number of unique connections: 3

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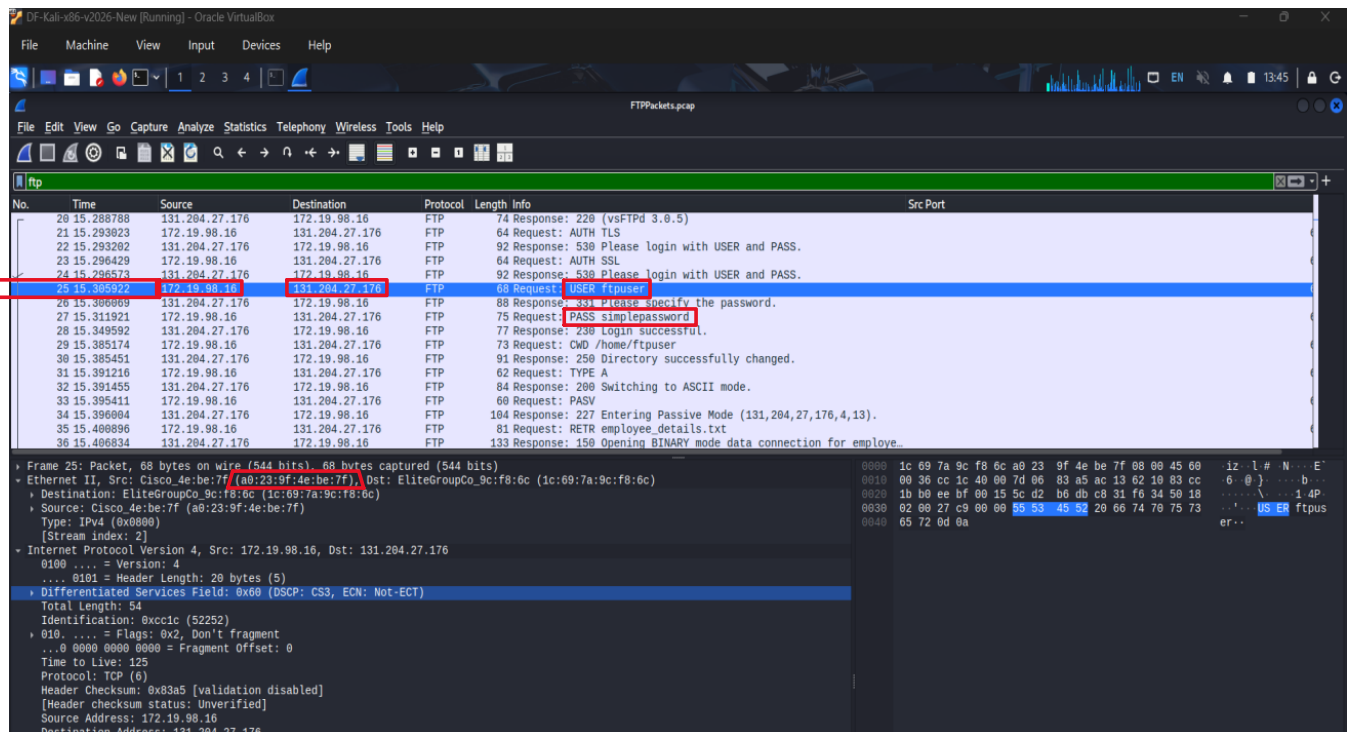
## 4. Authorized Login & Initial File Download

### 4.1 Authorized User Identification

- ftp\_username: ftpuser
- Login Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:48:51
- Source IP address: 172.19.98.16
- MAC address: a0:23:9f:4e:be:7f
- FTP response codes: 331 (user name okay, but needs password) and 230 (login successful)

Include annotated screenshots of:

- USER/PASS commands
- Successful login confirmation



FTP authentication traffic showing the USER and PASS commands for the authorized account ftpuser, coming from IP address 172.19.98.16

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The image shows a Wireshark packet capture of an FTP session. The filter is set to 'ftp.response\_code == 230'. The packet list shows several responses from the server to the client. The packet details pane for the selected packet (Frame 9) shows the following information:

- Frame 9: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits)
- Encapsulation type: Ethernet (I)
- Arrival Time: Nov 15, 2024 13:48:51.364503000 CST
- UTC Arrival Time: Nov 15, 2024 19:48:51.364503000 UTC
- Epoch Arrival Time: 1731700131.364503000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 35.634000 milliseconds]
- [Time since reference or first frame: 55.948000 milliseconds]
- Frame Number: 9
- Frame Length: 77 bytes (616 bits)
- Capture Length: 77 bytes (616 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:tcp:ftp]
- Character encoding: ASCII (0)
- [Coloring Rule Name: TCP]
- [Coloring Rule String: tcp]
- Ethernet II, Src: EliteGroupCo\_9c:f0:6c (1c:69:7a:9c:f0:6c), Dst: Cisco\_9f:f0:1b (00:00:0c:9f:f0:1b)
- Internet Protocol Version 4, Src: 131.204.27.176, Dst: 172.19.98.16
- Transmission Control Protocol, Src Port: 21, Dst Port: 61117, Seq: 131, Ack: 56, Len: 23
- File Transfer Protocol (FTP)
- [Current working directory: ]

| No.  | Time        | Source         | Destination    | Protocol | Length | Info                            | Src Port |
|------|-------------|----------------|----------------|----------|--------|---------------------------------|----------|
| 5    | 3.3556945   | 131.204.27.176 | 172.19.98.16   | FTP      | 77     | Response: 230 Login successful. | 21       |
| 26   | 15.349592   | 131.204.27.176 | 172.19.98.16   | FTP      | 77     | Response: 230 Login successful. | 21       |
| 47   | 233.956797  | 131.204.27.176 | 172.19.98.16   | FTP      | 77     | Response: 230 Login successful. | 21       |
| 74   | 294.938461  | 131.204.27.176 | 172.19.201.228 | FTP      | 77     | Response: 230 Login successful. | 21       |
| 879  | 1264.787286 | 131.204.27.176 | 172.19.201.228 | FTP      | 77     | Response: 230 Login successful. | 21       |
| 903  | 1468.629147 | 131.204.27.176 | 131.204.27.177 | FTP      | 89     | Response: 230 Login successful. | 21       |
| 1054 | 1761.350872 | 131.204.27.176 | 131.204.27.177 | FTP      | 89     | Response: 230 Login successful. | 21       |
| 1216 | 2168.551824 | 131.204.27.176 | 131.204.27.177 | FTP      | 89     | Response: 230 Login successful. | 21       |

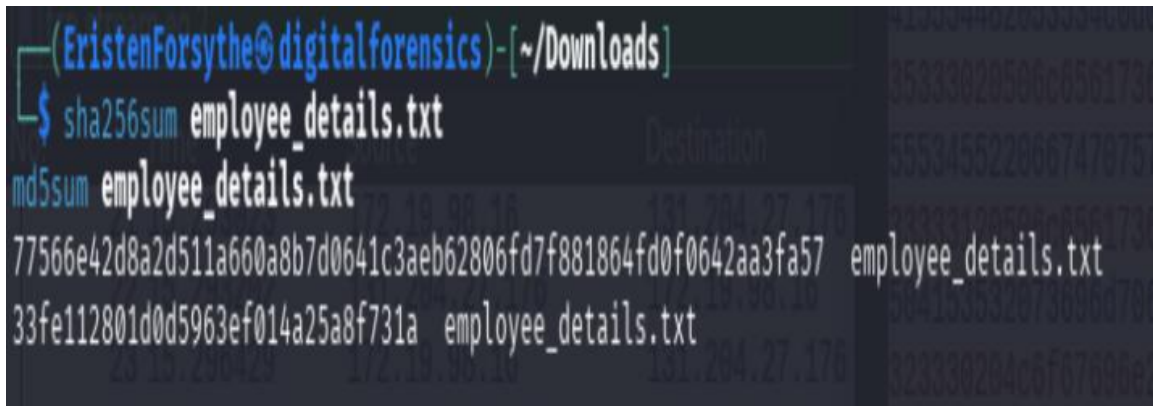
FTP server response 230 Login successful confirming authentication of the user ftpuser

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### 4.2 Initial File Download

- File name: employee\_details.txt
- File size (bytes): 485 bytes
- Transfer confirmation (FTP 150/226 responses): 150 (opening data connection) and 226 (transfer complete)
- File reconstruction steps:
  - Located the FTP command RETR employee\_details.txt in wireshark.
  - Right-clicked the packet and selected Follow -> TCP Stream.
  - Changed the stream view to raw data.
  - Saved the reconstructed file as employee\_details.txt.
- Hash values (SHA-256 and MD5):
  - SHA-256:  
77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57
  - MD5: 33fe112801d0d5963ef014a25a8f731a

A terminal window screenshot from a Kali Linux system. The prompt is (EristenForsythe@digitalforensics) - [~/Downloads]. The user has entered two commands: sha256sum employee\_details.txt and md5sum employee\_details.txt. The output shows the SHA-256 hash 77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57 and the MD5 hash 33fe112801d0d5963ef014a25a8f731a, both followed by the filename employee\_details.txt.

```
(EristenForsythe@digitalforensics) - [~/Downloads]  
$ sha256sum employee_details.txt  
md5sum employee_details.txt  
77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57 employee_details.txt  
33fe112801d0d5963ef014a25a8f731a employee_details.txt
```

Kali Linux hash verification for recovered employee\_details.txt

Recovered file contents included employee details and FTP credentials, including johnsmith/pass1234 and janedoe/1234abcd.

The recovered file contained employee personal information and plaintext FTP credentials. These credentials directly correspond to the later observed logins for the users johnsmith and janedoe in the packet capture.



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The image shows a Wireshark packet capture of an FTP session. The filter bar at the top is set to `ftp.request.command == "RETR"`. The packet list shows several FTP requests, with the selected packet (No. 35) being a RETR request for `employee_details.txt` from source `172.19.98.16` to destination `131.204.27.176`. The packet details pane shows the structure of the FTP request, including the current working directory `/home/ftpuser` and the command response frames.

| No.  | Time        | Source         | Destination    | Protocol | Length | Info                               |
|------|-------------|----------------|----------------|----------|--------|------------------------------------|
| 35   | 15.400896   | 172.19.98.16   | 131.204.27.176 | FTP      | 81     | Request: RETR employee_details.txt |
| 229  | 301.458302  | 172.19.201.228 | 131.204.27.176 | FTP      | 75     | Request: RETR statement2.pdf       |
| 922  | 1584.600071 | 131.204.27.177 | 131.204.27.176 | FTP      | 83     | Request: RETR check1.jpg           |
| 1004 | 1598.667621 | 131.204.27.177 | 131.204.27.176 | FTP      | 87     | Request: RETR statement1.pdf       |
| 1073 | 1784.950658 | 131.204.27.177 | 131.204.27.176 | FTP      | 86     | Request: RETR documents.zip        |

Frame 35: Packet, 81 bytes on wire (648 bits), 81 bytes captured  
Ethernet II, Src: Cisco\_4e:be:7f (a0:23:9f:4e:be:7f), Dst: Elite  
Internet Protocol Version 4, Src: 172.19.98.16, Dst: 131.204.27.176  
Transmission Control Protocol, Src Port: 61119, Dst Port: 21, Seq  
File Transfer Protocol (FTP)  
[Current working directory: /home/ftpuser]  
[Command response frames: 1]  
[Command response bytes: 485]  
[Command response first frame: 37]  
[Command response last frame: 37]  
[Setup frame: 34]

FTP RETR command showing the authorized user ftpuser requesting the file `employee_details.txt` from the FTP server `131.204.27.176`

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The image shows a Wireshark packet capture of an FTP session. The top pane displays a list of packets, with packet 1226 selected. The middle pane shows the details of the selected packet, which is an FTP response with code 150. The bottom pane shows the raw packet data in hexadecimal and ASCII. The packet list includes the following entries:

| No.  | Time     | Source         | Destination    | Protocol | Length | Info                                                                                    |
|------|----------|----------------|----------------|----------|--------|-----------------------------------------------------------------------------------------|
| 1226 | 22.72    | 131.204.27.176 | 131.204.27.177 | FTP      | 105    | Response: 150 Here comes the directory listing.                                         |
| 1269 | 22.90    | 131.204.27.176 | 131.204.27.177 | FTP      | 105    | Response: 150 Here comes the directory listing.                                         |
| 85   | 295.131  | 131.204.27.176 | 172.19.201.228 | FTP      | 76     | Response: 150 Ok to send data.                                                          |
| 1234 | 2198.194 | 131.204.27.176 | 131.204.27.177 | FTP      | 88     | Response: 150 Ok to send data.                                                          |
| 923  | 1584.686 | 131.204.27.176 | 131.204.27.177 | FTP      | 138    | Response: 150 Opening BINARY mode data connection for check1.jpg (184429 bytes).        |
| 1674 | 1784.951 | 131.204.27.176 | 131.204.27.177 | FTP      | 141    | Response: 150 Opening BINARY mode data connection for documents.zip (188768 bytes).     |
| 36   | 15.406   | 131.204.27.176 | 172.19.98.16   | FTP      | 133    | Response: 150 Opening BINARY mode data connection for employee_details.txt (485 bytes). |
| 1805 | 1598.668 | 131.204.27.176 | 131.204.27.177 | FTP      | 141    | Response: 150 Opening BINARY mode data connection for statement1.pdf (55871 bytes).     |
| 230  | 381.475  | 131.204.27.176 | 172.19.201.228 | FTP      | 130    | Response: 150 Opening BINARY mode data connection for statement2.pdf (885664 bytes).    |
| 19   | 0.147    | 131.204.27.176 | 172.19.98.16   | FTP      | 78     | Response: 226 Directory send OK.                                                        |
| 63   | 234.171  | 131.204.27.176 | 172.19.201.228 | FTP      | 78     | Response: 226 Directory send OK.                                                        |
| 226  | 295.258  | 131.204.27.176 | 172.19.201.228 | FTP      | 78     | Response: 226 Directory send OK.                                                        |
| 895  | 1264.988 | 131.204.27.176 | 172.19.201.228 | FTP      | 78     | Response: 226 Directory send OK.                                                        |
| 915  | 1475.844 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Directory send OK.                                                        |
| 1066 | 1764.698 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Directory send OK.                                                        |
| 1228 | 2173.398 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Directory send OK.                                                        |
| 1271 | 2290.398 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Directory send OK.                                                        |
| 38   | 15.412   | 131.204.27.176 | 172.19.98.16   | FTP      | 78     | Response: 226 Transfer complete.                                                        |
| 228  | 295.194  | 131.204.27.176 | 172.19.201.228 | FTP      | 78     | Response: 226 Transfer complete.                                                        |
| 858  | 381.673  | 131.204.27.176 | 172.19.201.228 | FTP      | 78     | Response: 226 Transfer complete.                                                        |
| 997  | 1584.686 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Transfer complete.                                                        |
| 1645 | 1598.673 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Transfer complete.                                                        |
| 1206 | 1784.967 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Transfer complete.                                                        |
| 1263 | 2198.198 | 131.204.27.176 | 131.204.27.177 | FTP      | 90     | Response: 226 Transfer complete.                                                        |

The details pane for packet 1226 shows the following information:

- Frame 1226: Packet, 105 bytes on wire (840 bits), 105 bytes captured (840 bits)
- Ethernet II, Src: EliteGroupCo.9c:f8:6c (1c:69:7a:9c:f8:6c), Dst: EliteGroupCo.9c:f9:06 (1c:69:7a:9c:f9:06)
- Internet Protocol Version 4, Src: 131.204.27.176, Dst: 131.204.27.177
- Transmission Control Protocol, Src Port: 21, Dst Port: 54488, Seq: 224, Ack: 54, Len: 39
- File Transfer Protocol (FTP)
  - 150 Here comes the directory listing.\r\n
  - Response code: File status okay; about to open data connection (150)
  - Response arg: Here comes the directory listing.
  - [Current working directory: ]

The raw packet data pane shows the hexadecimal and ASCII representation of the packet data.

FTP response codes 150 and 226 confirming the successful transfer of employee\_details.txt from the FTP server

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### 5. Unauthorized Access Enumeration

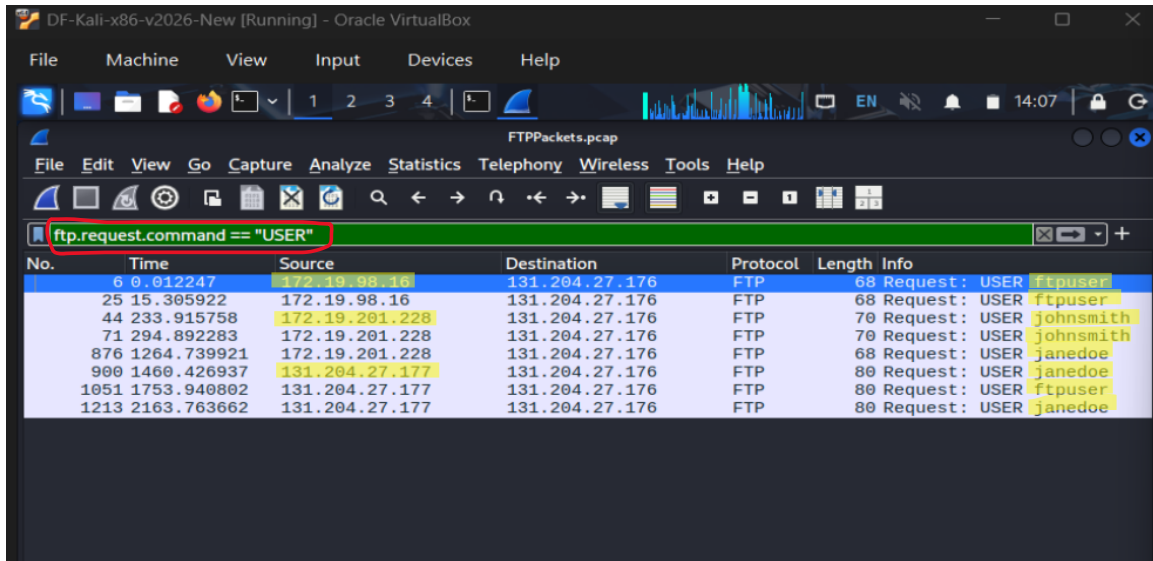
- Username: johnsmith
  - IP address: 172.19.201.228
  - MAC address (if observable): a0:23:9f:4e:be:7f
  - Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:52:36
  - FTP response codes: 331 (Username is okay, but needs password) and 230 (login successful)
  - Login Attempt: Successful
- 
- Username: janedoe
  - IP address: 131.204.27.177
  - MAC address (if observable): a0:23:9f:4e:be:7f
  - Timestamp (DD-MM-YYYY HH:MM:SS): 15-11-2024 13:54:16
  - FTP response codes: 331 (Username is okay, but needs password) and 230 (Login successful)
  - Login Attempt: Successful

| Username  | Source IP      | MAC Address       | Login Timestamp     | Assessment                                                 |
|-----------|----------------|-------------------|---------------------|------------------------------------------------------------|
| ftpuser   | 172.19.98.16   | a0:23:9f:4e:be:7f | 15-11-2024 13:48:51 | Authorized initial login                                   |
| ftpuser   | 172.19.98.16   | a0:23:9f:4e:be:7f | 15-11-2024 14:02:15 | Later reuse of authorized account                          |
| johnsmith | 172.19.201.228 | a0:23:9f:4e:be:7f | 15-11-2024 13:52:55 | Unauthorized – credentials exposed in employee_details.txt |
| johnsmith | 172.19.201.228 | a0:23:9f:4e:be:7f | 15-11-2024 14:04:34 | Unauthorized repeat login                                  |
| janedoe   | 131.204.27.177 | 1c:69:7a:9c:f9:06 | 15-11-2024 13:56:40 | Unauthorized – credentials exposed in employee_details.txt |
| janedoe   | 131.204.27.177 | 1c:69:7a:9c:f9:06 | 15-11-2024 14:18:06 | Unauthorized repeat login                                  |

Both unauthorized accounts correspond to credentials contained within employee\_details.txt, indicating that the credential file downloaded by ftpuser was likely used to authenticate subsequent unauthorized sessions.

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The image shows a Wireshark packet capture window titled 'FTPackets.pcap'. The filter bar at the top contains the expression 'ftp.request.command == "USER"', which is highlighted with a red rectangle. Below the filter bar, a table of captured packets is displayed, showing only those where the FTP command is 'USER'. The table has columns for No., Time, Source, Destination, Protocol, Length, and Info. The 'Info' column shows the request details, including the username being used for each login attempt.

| No.  | Time        | Source         | Destination    | Protocol | Length | Info                    |
|------|-------------|----------------|----------------|----------|--------|-------------------------|
| 6    | 0.012247    | 172.19.98.16   | 131.204.27.176 | FTP      | 68     | Request: USER ftpuser   |
| 25   | 15.305922   | 172.19.98.16   | 131.204.27.176 | FTP      | 68     | Request: USER ftpuser   |
| 44   | 233.915758  | 172.19.201.228 | 131.204.27.176 | FTP      | 70     | Request: USER johnsmith |
| 71   | 294.892283  | 172.19.201.228 | 131.204.27.176 | FTP      | 70     | Request: USER johnsmith |
| 876  | 1264.739921 | 172.19.201.228 | 131.204.27.176 | FTP      | 68     | Request: USER janedoe   |
| 900  | 1460.426937 | 131.204.27.177 | 131.204.27.176 | FTP      | 80     | Request: USER janedoe   |
| 1051 | 1753.940802 | 131.204.27.177 | 131.204.27.176 | FTP      | 80     | Request: USER ftpuser   |
| 1213 | 2163.763662 | 131.204.27.177 | 131.204.27.176 | FTP      | 80     | Request: USER janedoe   |

USER command filter showing all usernames observed in the FTP control channel above

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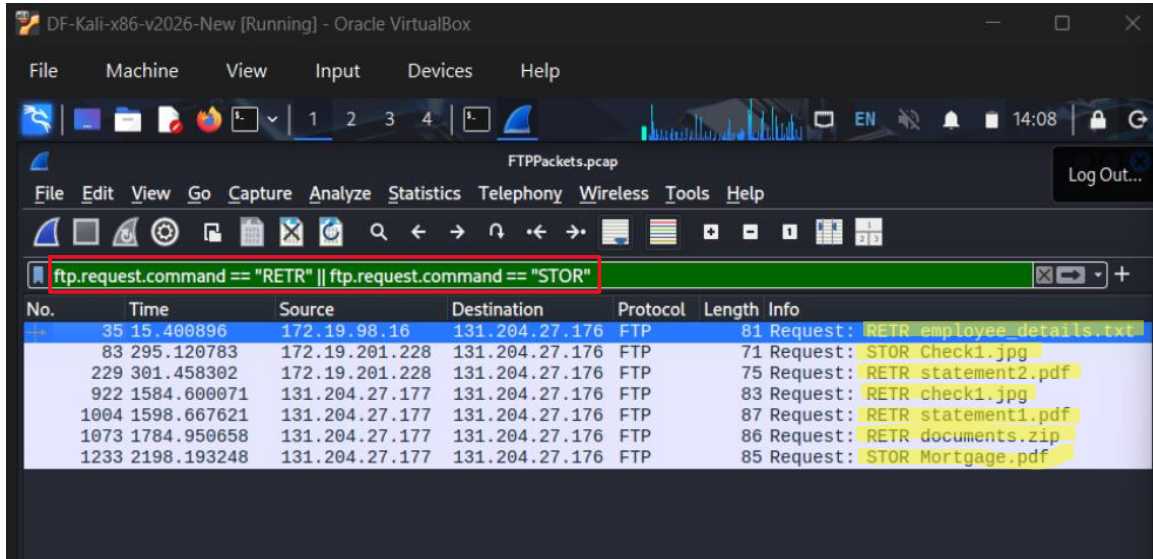
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### 6. File Activity Per Login

| Username  | IP Address     | Action   | File Name            | Size (bytes) | Integrity Notes                                                                  |
|-----------|----------------|----------|----------------------|--------------|----------------------------------------------------------------------------------|
| ftpuser   | 172.19.98.16   | Download | employee_details.txt | 485          | Credential leak file containing usernames, passwords, and employee personal data |
| johnsmith | 172.19.201.228 | Upload   | Check1.jpg           | 187392       | Completed check image; differs from check1.jpg template                          |
| johnsmith | 172.19.201.228 | Download | Statement2.pdf       | 47104        | Sample bank statement template                                                   |
| janedoe   | 131.204.27.177 | Download | check1.jpg           | 104448       | Sample/annotated check template                                                  |
| janedoe   | 131.204.27.177 | Download | Statement1.pdf       | 55296        | Realistic bank statement document                                                |
| janedoe   | 131.204.27.177 | Download | Documents.zip        | 189440       | Archive containing identity-related images (SSN card and birth record)           |
| janedoe   | 131.204.27.177 | Upload   | Mortgage.pdf         | 39936        | Blank mortgage approval template                                                 |

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RETR/STOR filter showing all FTP file-transfer requests in the capture above

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### 7. Recovery of All Transferred Files

| File Name            | File Size (bytes) | SHA-256                                                            | MD5                              | Export Method                                 |
|----------------------|-------------------|--------------------------------------------------------------------|----------------------------------|-----------------------------------------------|
| Employee_details.txt | 485               | 77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57   | 33fe112801d0d5963ef014a25a8f731a | Wireshark → Follow TCP Stream → Save Raw Data |
| Check1.jpg           | 187392            | f87999f63685b4dd0dc5193b2ce199c8eb7f0cf57b0bdb4c1924b53df7eed74    | 828a6aaf7d03425b709d1c934956e215 | Wireshark → Export Objects → FTP              |
| check1.jpg           | 104448            | 5031d7d419f8dc734ef0755238903b46bbcbf00148f20ac28f40ca2626032fc3   | 7127fbc910bba91881e4f92d9a60a3a4 | Wireshark → Export Objects → FTP              |
| Statement1.pdf       | 55296             | 6196dd97ba8071953c61e667dbdfaa1d6007025d85e150e75d29e82af9f241d9   | 5d73628bbc55be7a97b165793ff89a29 | Wireshark → Export Objects → FTP              |
| Statement2.pdf       | 47104             | 53cb0d6fcea0691c567f874309df843df7d7386842a54b3097825fc3ddeb235c   | a347008364d3a9c640ee20fb098113c3 | Wireshark → Export Objects → FTP              |
| Documents.zip        | 189440            | 5cedf6a3089171e677adc7470f97b396d0f42168e95aa0925f0de5a0bf0ac722   | 7e256228ed332013dc8857bc4b735b88 | Wireshark → Export Objects → FTP              |
| Mortgage.pdf         | 39936             | 944393f110e37278dfef31bb9bd250252e681593bfbfd773a45bab48faa023fab7 | 1f4027a6add0e65c7ccd054a4fc066b  | Wireshark → Export Objects → FTP              |

All reconstructed files listed above are in the submission zip file 'recovered\_files.zip'.



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```
DF-Kali-x86-v2026-New [Running] - Oracle VirtualBox
File Machine View Input Devices Help

EristenForsythe@digitalforensics: ~/Downloads/Project3
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$ ls
check1.jpg Check1.jpg documents.zip employee_details.txt Mortgage.pdf statement1.pdf statement2.pdf

(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$ sha256sum check1.jpg
sha256sum documents.zip
sha256sum employee_details.txt
sha256sum Mortgage.pdf
sha256sum statement1.pdf
sha256sum statement2.pdf

(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$ md5sum check1.jpg
md5sum Check1.jpg
md5sum documents.zip
md5sum employee_details.txt
md5sum Mortgage.pdf
md5sum statement1.pdf
md5sum statement2.pdf

5031d7d419f8dc734ef0755238903b46bbcbf00148f20ac28f40ca2626032fc3 check1.jpg
f87999f6368544ddd0dc5193b2ce199c8eb7f0cf57b0bdb4c1924b53df7eed74 Check1.jpg
5cedf6a3089171e677adc7470f97b396d0f42168e95aa0925f0de5a0bf0ac722 documents.zip
439d16e6078203e471acc0c751f08c8e6b5faba91a28c598069152c281873617 employee_details.txt
944393f110e37278dfb31bb9bd250252e681593bfd773a45bab48faa023fab7 Mortgage.pdf
53cb0dd6fcea0691c567f874309df843df7d7386842a5443907825fc3ddeb235c statement1.pdf
6196dd97ba8071953c61e667dbdffa1d6007025d85e150e75d29e82af9f241d9 statement2.pdf
7127fbc910bba91881e4f92d9a60a3a4 check1.jpg
828a6aaf7d03425b709d1c934956e215 Check1.jpg
7e256228ed332013dc8857bc4b735b88 documents.zip
a02a61ff501d8e45d5b80142682e361a employee_details.txt
1f4027a6add04e6547ccd054a4fc066b Mortgage.pdf
5d73628bbc55be7a97b165793ff89a29 statement1.pdf
a347008364d3a9c640ee20fb098113c3 statement2.pdf

(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$
```

SHA-256 and MD5 hashes for all recovered files above

```
(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$ ls -lh
total 1.4M
-rw-r--r-- 1 EristenForsythe EristenForsythe 102K Mar  9 14:10 check1.jpg
-rw-r--r-- 1 EristenForsythe EristenForsythe 183K Mar  9 14:10 Check1.jpg
-rw-r--r-- 1 EristenForsythe EristenForsythe 185K Mar  9 14:10 documents.zip
-rw-r--r-- 1 EristenForsythe EristenForsythe  485 Mar  9 14:10 employee_details.txt
-rw-r--r-- 1 EristenForsythe EristenForsythe  39K Mar  9 14:10 Mortgage.pdf
-rw-r--r-- 1 EristenForsythe EristenForsythe  54K Mar  9 14:10 statement1.pdf
-rw-r--r-- 1 EristenForsythe EristenForsythe  846K Mar  9 14:10 statement2.pdf

(EristenForsythe@digitalforensics)-[~/Downloads/Project3]
$
```

ls -lh output showing the recovered files in the Project3 directory

Files were reconstructed from the packet capture using two methods in Wireshark.



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The file employee\_details.txt was recovered by locating the RETR command and using

Follow → TCP Stream to save the raw file data.

The remaining transferred files were recovered using

File → Export Objects → FTP in Wireshark, which reconstructs files transferred through the FTP data channel.

---

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### 8. Consolidated Findings Table (Required)

| Username  | IP Address     | MAC Address       | Login Timestamp (DD-MM-YYYY HH:M M:SS) | Action   | File Name + Ext      | File Size (Bytes) | File Hash                                                                |
|-----------|----------------|-------------------|----------------------------------------|----------|----------------------|-------------------|--------------------------------------------------------------------------|
| ftpuser   | 172.19.98.16   | a0:23:9f:4e:be:7f | 15-11-2024 13:48:51                    | Download | Employee_details.txt | 485               | SHA256: 77566e42d8a2d511a660a8b7d0641c3aeb62806fd7f881864fd0f0642aa3fa57 |
| johnsmith | 172.19.201.228 | a0:23:9f:4e:be:7f | 15-11-2024 13:52:36                    | Upload   | Check1.jpg           | 187392            | SHA256: f87999f63685b4ddd0dc5193b2ce199c8eb7f0cf57b0bdb4c1924b53df7eed74 |
| johnsmith | 172.19.201.228 | a0:23:9f:4e:be:7f | 15-11-2024 13:52:36                    | Download | Statement2.pdf       | 47104             | SHA256: 53cb0d6fcea0691c567f874309df843df7d7386842a54b3097825fc3ddeb235c |
| janedoe   | 131.204.27.177 | a0:23:9f:4e:be:7f | 15-11-2024 13:54:16                    | Download | check1.jpg           | 104448            | SHA256: 5031d7d419f8dc734ef0755238903b46bbcbf00148f20ac28f40ca2626032fc3 |
| janedoe   | 131.204.27.177 | a0:23:9f:4e:be:7f | 15-11-2024 13:54:16                    | Download | Statement1.pdf       | 55296             | SHA256: 6196dd97ba8071953c61e667dbdfaa1d6007025d85e150e75                |

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|         |                |                   |                        |          |               |        |                                                                             |
|---------|----------------|-------------------|------------------------|----------|---------------|--------|-----------------------------------------------------------------------------|
|         |                |                   |                        |          |               |        | d29e82af9f241d9                                                             |
| janedoe | 131.204.27.177 | a0:23:9f:4e:be:7f | 15-11-2024<br>13:54:16 | Download | Documents.zip | 189440 | SHA256:<br>5cedf6a3089171e677adc7470f97b396d0f42168e95aa0925f0de5a0bf0ac722 |
| janedoe | 131.204.27.177 | a0:23:9f:4e:be:7f | 15-11-2024<br>13:54:16 | Upload   | Mortgage.pdf  | 39936  | SHA256:<br>944393f110e37278dfeb31bb9bd250252e681593bfd773a45bab48faa023fab7 |

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### 9. Conclusion

The analysis of the FTTPackets.pcap file showed that the authorized user ftpuser initially logged into the FTP server and downloaded the file employee\_details.txt. This file had employee usernames, passwords, and personal information. Shortly after the download the credentials contained in the file were used to authenticate additional FTP sessions under the accounts johnsmith and janedoe.

Once authenticated, these accounts performed multiple file transfers involving images, PDFs, and a ZIP archive. The transferred files included financial documents such as bank statements, check images, and a mortgage template, as well as an archive containing identity related images like a social security card and birth certificate. All transferred files were successfully reconstructed from the packet capture and verified using SHA-256 and MD5 hashes.

The activity suggests that the exposed credentials enabled unauthorized access to the FTP server and allowed the attackers to exchange files containing personal identity information and financial documents. The materials could potentially be used for identity theft or financial fraud.

Overall the packet capture provided clear forensic evidence of credential exposure, unauthorized authentication, and subsequent file transfers. The recovered files and associated hash values allow the actions of each user session to be verified and reconstructed.